Minimisation Algorithm

2026-04-17

Introduction

Minimisation, introduced by Taves (1974) and refined by Pocock and Simon (1975), is a covariate-adaptive allocation procedure that assigns each new participant to the treatment arm that prospectively makes the distribution of pre-specified prognostic covariates most balanced across arms. Unlike stratified block randomisation, which requires a separate randomisation list for every stratum and quickly becomes impractical when more than two or three covariates need balancing, minimisation handles many prognostic factors simultaneously in a single allocation framework. It is now widely used in trials with several important baseline covariates and small-to-moderate sample sizes, where stratified randomisation would create too many empty strata to be useful.

Prerequisites

A working understanding of simple, block, and stratified randomisation; balance metrics for cross-tabulated baseline covariates; and the regulatory framework around covariate-adaptive allocation.

Theory

For each candidate arm assignment, the algorithm computes a balance score — typically the sum across covariates of marginal imbalances that would result from that assignment. The new participant is allocated to the balance-minimising arm with high probability p (commonly 0.8 or 0.9), and to the other arm with probability $1 - p$ to preserve a degree of allocation unpredictability. The probabilistic element matters: a deterministic minimisation that always chooses the balance-minimising arm becomes predictable to investigators who know the algorithm and the previous assignments, compromising allocation concealment.

Assumptions

The covariates to be balanced are pre-specified in the protocol, allocation is performed centrally (typically through an interactive web-response system) rather than manually, and the trial's analysis model includes all minimisation covariates as fixed effects to preserve valid inference under randomisation theory.

R Implementation

```r
library(Minirand)

set.seed(2026)
n <- 60
covmat <- data.frame(
  centre = sample(c("A", "B", "C"), n, replace = TRUE),
  sex    = sample(c("M", "F"), n, replace = TRUE),
  age_g  = sample(c("young", "old"), n, replace = TRUE)
)

res <- character(n)
for (j in 1:n) {
```

```r
  res[j] <- Minirand(covmat = covmat, covwt = rep(1, 3),
                     ntrt = 2, trtseq = c("A", "B"),
                     ratio = c(1, 1),
                     p = 0.9, j = j, result = res)
}

table(trt = res, centre = covmat$centre)
table(trt = res, sex = covmat$sex)
```

Output & Results

`Minirand()` allocates each subject sequentially based on the prior allocation history and the new subject's covariate profile. Cross-tabulations of treatment by each covariate show approximately equal arm counts within every covariate level — the design's primary objective — even when no single covariate combination has many subjects.

Interpretation

A reporting sentence: "Treatment allocation used Pocock-Simon minimisation balancing on three pre-specified covariates (centre, sex, age category), each with equal weight; the random component used probability 0.9 of allocation to the balance-minimising arm. The trial's analysis model retained centre, sex, and age category as fixed-effect covariates to preserve valid inference under minimisation; in the final 60-subject sample, the maximum marginal arm imbalance on any covariate was 1 subject." Always state the random-component probability and the analysis-model covariates.

Practical Tips

- Always analyse the trial with the minimisation covariates as fixed-effect adjustments in the primary analysis model; omitting them violates the randomisation-inference framework that minimisation relies on, and the resulting standard errors are typically too small.
- Pre-specify the covariates, their weights, and the random-component probability p in the protocol and SAP; adding covariates post hoc defeats minimisation's protective role and is disallowed by most regulators.
- Commercial IWRS systems are required for robust minimisation in any non-trivial trial; manual implementation is error-prone, especially as the trial grows, and a single manual error can compromise allocation concealment for the entire study.
- FDA and EMA accept minimisation when the method, covariates, and analysis model are pre-specified; the regulatory concern about covariate-adaptive allocation is largely addressed by transparent documentation and analysis-model adjustment.
- Minimisation is less transparent than stratified block randomisation — investigators cannot reproduce the allocation list from a simple description — so the protocol should describe the algorithm carefully, including the balance metric, weights, and probability parameter.
- For trials with very few prognostic covariates and moderate sample size, stratified block randomisation is often preferable because of its operational simplicity; minimisation's advantage grows with the number of covariates and with smaller per-stratum sample sizes.

R Packages Used

`Minirand` for canonical Pocock-Simon minimisation with arbitrary weights and ratio support; `randomizeR` for an alternative interface integrated with broader randomisation simulation; `bcrm` for biased-coin and minimisation alternatives; `RandomizationLogic` for full IWRS-style simulation including audit trail; `Mediana` for trial-design simulation with minimisation-allocation strategies.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale